

October 2, 2026

The Editors
Information Processing & Management

Re: Submission of a Full Length Article — “Sieve: Budgeted Question-Conditioned Information Selection for Training-Free Long-Context Compression”

Dear Editors,

Please consider the manuscript referenced above for publication in *Information Processing & Management* as a **Full Length Article**. I am the sole author of the manuscript and serve as its corresponding author.

I would like to be transparent about this manuscript’s history. An earlier version was previously submitted to *IP&M* and was declined, with the editorial assessment that the results appeared premature. The present manuscript is a substantially reframed and extended version, prepared specifically to address that assessment: the contribution is now stated as a problem formulation rather than a single system, the evidence base has been scaled and deepened with three new experiment families, and the two protocols that previously rested on the thinnest samples have been re-run at size. A summary of what changed follows; I am submitting it as a new submission and leave the editorial decision on how to handle the history entirely to you.

The manuscript formulates long-context compression as *budgeted question-conditioned information selection*: given a fixed word budget, choose the subset of sentences that jointly maximizes question-relevance coverage, positional reliability, and diversity, with the selector’s own computational cost treated as a first-class constraint rather than an afterthought. This formulation, to our knowledge absent from prior work even where its ingredients appear as isolated heuristics, is stated formally, and a twelve-selector design space table locates every prior system along seven dimensions (question conditioning, positional awareness, redundancy control, model-free inference, training-free operation, sentence-level intact units, explicit budget with measured cost). The unoccupied corner the table exposes is the one this paper measures: a selector that is simultaneously question-conditioned, position-aware, redundancy-controlling, and entirely model-free.

The vehicle of the study is *Sieve*, the instantiation of that formulation in this corner: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under the explicit budget, with selected sentences emitted in their original order. Its preprocessing is a linear-time pass over the context, its budgeted selection is $O(k n \bar{s})$, and the full selector runs in tens of milliseconds on one CPU core (50.9–54.3 ms at a 6,000-word context), with no model parameters, no training, and no dependency beyond NumPy.

The evaluation is organized as a three-way quality–compression–cost frontier study, and it is the part that has grown the most since the earlier version. First, the reader-free evidence-recall protocol now spans three evidence-annotated benchmarks — QASPER, HotpotQA

(distractor setting), and 2WikiMultihopQA — 1,600 records in total: *Sieve* retains more gold evidence than every question-agnostic baseline on all three datasets (margins over head truncation of 12.8, 44.8, and 22.8 recall points at $4\times$), while BM25 sentence ranking, the strongest question-conditioned model-free comparator, exceeds the full selector by 6.8 points on QASPER — a result the manuscript reports as measured and reads through its own ablations as a structural trade of raw recall for positional reliability. Second, the end-task protocol is new since the earlier submission and is now run at size: seven selection methods (full-context, head, random, stride, TextRank, BM25, *Sieve*) at $n=100$ each under a production LLM reader at the $4\times$ operating point, with paired bootstrap intervals throughout. The selector-level ordering reappears end to end: the BM25 reference retains 76.0% of full-context F1 against 61.8% for *Sieve* (paired gap 4.8 points, interval excluding zero), while *Sieve* remains statistically matched with truncation and the question-blind floor (paired gap +4.0, interval crossing zero) — the same trade-off, now consistent at both the selection and the answer layer. Third, a full configuration–budget–dataset ablation matrix isolates each component’s contribution: the relevance term alone is a near-oracle for evidence recovery (43.6% versus 36.2% for the full configuration on QASPER), the positional prior’s price is a dataset-stable six-to-seven paired points, and all five selector configurations run in the same milliseconds band, so the trade is paid in information units rather than in compute. Fourth, the reader-side positional profile that motivates the U-shaped prior is no longer borrowed from cited work: it is re-measured under our own reader at $n=50$ paired records across five evidence positions, and reported flat at this context scale (median $\sim 1,000$ words), which the manuscript reports as a finding about scope: the prior remains a selector-level design choice whose cost the ablation prices. Fifth, a GPT-2 perplexity gate, measured on the same hardware as every other method, reaches 28.7% evidence recall at $255\text{--}320\times$ *Sieve*’s measured cost; and the correlation between evidence recall and end-task answer F1 is re-estimated at the new sample size (pooled Pearson 0.22 over 600 pairs), so the two protocols are reported separately throughout.

I believe the manuscript fits the scope of *Information Processing & Management*. It is a quantitative study of question-conditioned information selection: it builds directly on classical information-retrieval machinery (IDF weighting, BM25, maximal marginal relevance), evaluates selection quality with recall protocols over annotated evidence, connects selection decisions to the processing and cost of managing long contexts in information systems, and reports quality, compression, and compressor cost for every method so that practitioners have a measured basis for deciding when statistical selection suffices and when a hosted scorer earns its infrastructure.

The manuscript substantially extends an earlier conference-format study of the same method, as described in the Introduction: a three-dataset evaluation, the BM25 baseline, the same-hardware comparison against a neural perplexity gate, the formal layer (formulation, propositions, cost statements), the correlation study, the end-to-end seven-method protocol at $n=100$, the configuration–budget–dataset ablation matrix, and the reader-side positional profile.

I confirm that: the manuscript is original and has not been published previously in any form; it is not under consideration by any other journal; I have no known competing financial

interests or personal relationships that could have appeared to influence the work reported; and the work received no specific funding. The benchmarks used are public (QASPER, HotpotQA, 2WikiMultihopQA), and the released repository reproduces every table and figure from raw checkpoints.

Thank you for considering the manuscript. I look forward to the reviewers' comments.

Sincerely,

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